ANDROID DEVELOPMENT ROADMAP (JAVA + 20MCA243)

Legend:

Completed = [Done]

Partial = [Partial]

Not Started = [Todo]

University Syllabus = [Syllabus]

1. JAVA FUNDAMENTALS

- Variables [Done]
- Data Types [Done]
- Operators [Done]
- if-else [Done]
- switch [Done]
- Loops [Done]
- Arrays [Done]
- ArrayList [Done]
- Methods [Done]
- Classes & Objects [Done]
- OOP Concepts [Done]

2. ANDROID FUNDAMENTALS [Syllabus]

- Android Studio [Done]
- Project Structure [Done]
- AndroidManifest.xml [Done]
- Activities [Done]
- Activity Lifecycle [Done]
- Gradle [Done]
- Services [Todo]
- Broadcast Receivers [Todo]
- Content Providers [Todo]
- Views [Partial]
- Notification Windows [Todo]
- Intent Filters [Todo]

3. RESOURCE MANAGEMENT [Syllabus]

- R.java / Resources [Partial]
- Assets Folder [Todo]
- XML Layouts [Done]
- Drawable Resources [Partial]
- styles.xml [Todo]
- Themes [Todo]
- Selectors [Todo]
- Vector Drawables [Todo]

4. EMULATOR & DEBUGGING [Syllabus]

- Emulator (AVD) [Done]
- Emulator Profiles [Todo]
- Logcat [Partial]
- DDMS [Todo]

5. LAYOUTS [Syllabus]

- LinearLayout [Done]
- RelativeLayout [Done]

- ConstraintLayout [Done]
- ScrollView [Todo]
- LayoutParams [Done]
- dp/sp/px [Todo]
- Input Validation [Todo]

6. UI WIDGETS [Syllabus]
- TextView [Done]
- EditText [Done]
- Button [Done]
- ImageView [Todo]
- CheckBox [Done]
- RadioButton [Done]
- RadioGroup [Done]
- Switch [Todo]
- ToggleButton [Todo]
- Spinner [Todo]
- AutoCompleteTextView [Todo]
- ProgressBar [Todo]
- SeekBar [Todo]
- RatingBar [Todo]
- DatePicker [Todo]
- TimePicker [Todo]

7. EVENT HANDLING
- OnClickListener [Done]
- OnLongClickListener [Done]
- OnItemClickListener [Done]
- OnCheckedChangeListener [Todo]
- OnItemSelectedListener [Todo]

8. USER FEEDBACK [Syllabus]
- Toast [Done]
- Snackbar [Todo]
- AlertDialog [Todo]
- Popup Menu [Done]

9. INTENTS [Syllabus]
- Explicit Intent [Done]
- Implicit Intent [Todo]
- Passing Data [Done]
- Receiving Data [Done]
- Intent Filters [Todo]

10. MULTIPLE ACTIVITIES
- Navigation [Done]
- Back Stack [Todo]
- Activity Result [Todo]

11. LIST COMPONENTS [Syllabus]
- ListView [Done]
- ArrayAdapter [Done]
- Custom Adapter [Todo]
- RecyclerView [Todo]

- CardView [Todo]
12. MENUS [Syllabus]
- Options Menu [Done]
- Context Menu [Done]
- Popup Menu [Done]
- Toolbar [Done]
- XML vs Java Menu Creation [Partial]
13. DATA STORAGE [Syllabus]
- SharedPreferences [Todo]
- Preference XML [Todo]
- Internal Storage [Todo]
- SQLite [Todo]
- SQLiteOpenHelper [Todo]
- SQLiteDatabase [Todo]
- Cursor [Todo]
- Room Database [Todo]
14. FRAGMENTS & NAVIGATION [Syllabus]
- Fragments [Todo]
- Fragment Lifecycle [Todo]
- FragmentManager [Todo]
- Fragment Transactions [Todo]
- Fragment Communication [Todo]
- Navigation Drawer [Todo]
- Tab Layout [Todo]
15. DEVICE FEATURES [Syllabus]
- Camera [Todo]
- Gallery [Todo]
- Audio/Video [Todo]
- Runtime Permissions [Todo]
- GPS/Location [Todo]
- Sensors [Todo]
- Notifications [Todo]
16. CONTENT PROVIDERS [Syllabus]
- Contacts Provider [Todo]
- Browser Provider [Todo]
- Custom Content Provider [Todo]
17. NETWORKING
- JSON [Todo]
- REST APIs [Todo]
- Volley [Todo]
- Retrofit [Todo]
- Firebase [Todo]
18. ARCHITECTURE
- MVVM [Todo]
- ViewModel [Todo]
- LiveData [Todo]
- Navigation Component [Todo]
19. COMPLETE PROJECTS

- Student Management [Done]
- Notes App [Todo]
- To-Do App [Todo]
- Expense Tracker [Todo]
- Quiz App [Todo]
- Food Ordering App [Todo]